

Mazen Emam

Frontend Engineer | React.js · TypeScript | Incoming M.Sc. Student at FAU

Alexandria, Egypt | Relocating to Nürnberg, Germany in October 2026 | German: A0 (Beginner)

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[LinkedIn](#) | [GitHub](#) | [Medium](#) | [Portfolio](#)

PROFILE

Frontend engineer with 4 years shipping production React/TypeScript systems for manufacturing, threat-analytics, and recruitment platforms across MENA and Europe - including a production AI chat feature querying 300+ live sources, plus independent AI/LLM engineering across two self-directed projects. Relocating to Nürnberg for an M.Sc. in International Information Systems (FAU, Oct 2026), authorized to work in Germany as an international student, targeting Frontend and Software Engineer roles with substantial hours from October 2026, transitioning to full-time upon program completion.

EDUCATION

Friedrich-Alexander-Universität (FAU) · Erlangen, Germany

Starting Oct 2026

M.Sc. International Information Systems

[Information Technology Institute \(ITI\)](#)

Sep 2020 – Dec 2020

MEARN Track, 3-Month Diploma — bridge into software engineering

Alexandria University · Alexandria, Egypt

Oct 2013 – Jul 2017

B.Sc. Management Information Systems

WORK EXPERIENCE

[Siren Analytics](#) · Beirut, Lebanon

Jun 2025 – Jan 2026

Software Engineer | React.js · TypeScript · Material UI · React Query · Redux Toolkit · WebSocket · SockJS/STOMP · React Hook Form · Yup

- Led development of an AI chat module enabling users to query 300+ news sources conversationally, replacing manual content navigation across bulk-ingested datasets.
- Implemented role-based access control across a multi-tenant architecture, scoping feature and content visibility per tenant and enforcing user-level permissions platform-wide.
- Engineered a multi-language translation system with React Query caching and bulk translation hooks, supporting the company's expansion into the European market.
- Pioneered a real-time WebSocket notification system using SockJS/STOMP with throttled error handling, enabling instant cross-user threat flagging that navigates recipients directly to flagged content.
- Optimized performance by integrating React Compiler and leading application-wide code splitting, reducing unnecessary re-renders and improving bundle efficiency.
- Secured authentication by refactoring JWT expiry logic and eliminating race conditions in token refresh, preventing unauthorized access.
- Eliminated a class of runtime errors by establishing a TypeScript enforcement pipeline with strict ESLint rules and a pre-push hook, catching type inconsistencies before code integration.

[Garment IO](#) · Giza, Egypt

May 2023 – Jun 2025

Mid-Senior Frontend Engineer | React.js · TypeScript · Material UI · Redux · Sentry · Formik · Yup · AWS · GitHub Actions

- Introduced TypeScript and strict ESLint enforcement across a 6-year-old codebase spanning 10+ modules and 40+ pages via an incremental migration strategy, establishing type safety without disrupting active development.
- Modernized the build pipeline by migrating from Create React App to Vite, reducing production build time by 50% and improving local development speed by 5x.
- Designed an i18n system with real-time language switching across 4 languages, enabling product deployment across MENA and European markets.
- Implemented Sentry performance monitoring capturing Web Vitals and API latency bottlenecks, identifying production issues across 30+ manufacturing facilities.
- Built a QR code scanning system for line managers, embedding machine IDs into scannable codes and eliminating manual lookup across 30+ facilities.
- Resolved performance bottlenecks and memory leaks by converting sequential API calls to parallel execution, removing unused data fetching, and enforcing pagination on unbounded endpoints.

Frontend Engineer | React.js · TypeScript · Material UI · MobX · React Hook Form · Yup

- Established a reusable component library with RTL support for the company's recruitment platform, creating a consistent design system that reduced new feature development time by 50%.
- Accelerated app startup speed by 35% through image optimization, lazy loading, and efficient API integrations.

Akwad · Alexandria, Egypt

Oct 2021 – Jul 2022

Frontend Engineer | React.js · JavaScript · Bootstrap · Context API · React Hook Form

- Created a reusable form and grid component framework, standardizing development and reducing page build time by 90%.
- Overhauled the core data grid infrastructure, unifying pagination and search logic to eliminate inconsistent behavior and reduce bugs.

PROFESSIONAL DEVELOPMENT & RELOCATION

Germany Preparation

Feb 2026 – Oct 2026

- Continued developing and maintaining independent software projects, including Job Radar and Echo FM, while strengthening my frontend and full-stack development skills.
- Secured admission to the M.Sc. International Information Systems program at FAU; securing visa and finalizing relocation logistics ahead of program start.

PROJECTS

Job Radar — AI-Powered Job Dashboard[GitHub](#) – [Live](#) | *Next.js 14 · TypeScript · React · Tailwind CSS · Supabase (PostgreSQL) · Google Gemini API · GitHub Actions*

- Built the entire user-facing product in Next.js 14 and TypeScript, including a personalized job dashboard, a multi-step settings and onboarding flow, an application tracker, and a salary insights explorer.
- Designed the dashboard so every user sees results scored against their own skills, seniority preference, and keyword rules instead of a generic feed, giving users a fully personalized experience from the same dataset.
- Integrated Google's Gemini API to power AI-assisted job filtering and on-demand application strategy suggestions, including a five-model fallback chain so the feature stays reliable when a model hits a rate limit or error.
- Engineered a caching layer so returning users get instant dashboard loads, while data automatically recomputes in the background whenever the underlying job pool or a user's settings change, balancing perceived performance against compute cost.
- Implemented secure, role-based access end to end with Google OAuth and Supabase, including route-level middleware protection and an admin console, with safeguards that prevent privilege escalation through the API.

Echo FM — Autonomous AI Satirical Radio Station[GitHub](#) – [Live](#) – [Mirror](#) | *Python · GitHub Actions · Supabase · Webhooks · LLM · FFmpeg · Serverless Architecture*

- Designed and shipped a fully autonomous, serverless media pipeline: live news ingestion → LLM script generation → TTS synthesis → LUFS-mastered audio → automated YouTube publishing, running on a 6-hour CRON schedule with zero manual intervention.
- Integrated multiple LLM APIs (Gemini, Groq/Llama-4) with a custom Smart Router and quality validator; built a 3-tier TTS cascade enforcing vocal consistency across 13-segment broadcasts via a Unified Master Engine pattern.

TECHNOLOGIES

Proficient: React.js · TypeScript · JavaScript · Redux/Redux Toolkit · React Query/TanStack Query · Material-UI · Vite · HTML · CSS · SASS · Bootstrap · Git

Familiar: Next.js · Node.js · Express · MongoDB · GraphQL · WebSocket/STOMP · MobX · Sentry · Jira · Jest · Vitest · React Testing Library · AWS · Docker · Webpack

CERTIFICATES

[Udacity — React Nanodegree](#)

Jul 2022 – Aug 2022

[freeCodeCamp — JavaScript Algorithms and Data Structures](#)

Jun 2022 – Jul 2022

LANGUAGES

Arabic — Native | English — Fluent (C1) | German — Beginner (A0)